

The Extraction Skill: Drawing Penalties Belongs to the Player, Not His Line or the Referees

Introduction

When a player draws a penalty in hockey, his team gets a power play, and the official record charges only the offender. Analysts have known for a decade that drawing penalties repeats year to year. Three things have not been tested: whether the skill belongs to the player or to the five skaters on the ice, whether it survives a change of referees, and whether coaches reward it with ice time. It is the player's, it survives, and they do not.

Methods

This paper estimates each skater's rate of drawing even-strength minors from three NHL seasons of play-by-play (3,936 games), relative to what his ice time, position, opponents, score and team predict, pulled toward average in small samples. The estimate is fit on two seasons and tested on the third, against four questions. Does it beat the player's raw rate? Does crediting everyone on the ice rank the same players? Does the ranking hold across referee crews that share no officials? Does skating speed in NHL Edge tracking explain it? The paper also checks persistence over fifteen seasons, values a drawn penalty by what followed at the same score and time, and repeats the construction in the NBA, WNBA, and soccer's World Cups, Euros and women's club leagues.

Results

Prior skill predicts held-out draw totals at **0.72**, against 0.25 for ice time alone and 0.53 with context added. A player's raw prior rate predicts just as well (**0.72**): adjusting for circumstances changes almost nothing, because the rate was his all along. The ranking is positive in all fourteen season pairs (career reliability 0.89) and as stable across disjoint referee crews (0.54) as across a random split (0.55). A model crediting everyone on the ice ranks Panarin and Fiala first and correlates **0.09** with the individual ranking, led by Hathaway, Pachal and Tkachuk. Faster skaters draw more trips, hooks and holds (partial correlation 0.16 to 0.23) and no more roughing or cross-checks (0.04). Drawing is unrelated to scoring (0.07), and players who draw penalties tend to take them (0.50). A drawn minor is worth 0.017 wins; the best player, net of penalties taken, adds **0.3 wins** per season. At equal scoring, better drawers get slightly less ice time. In the NBA, WNBA and men's soccer the same method ranks Antetokounmpo, Wilson and Hazard first (reliability 0.64 to 0.94).

Conclusion

Penalty drawing is an individual talent; models that credit everyone on the ice measure something else. For a team, drawing can be had without taking: one player in six draws more than expected and takes fewer (McDavid, Ehlers, Skinner), while the agitator who draws the most gives much of it back. Ice time does not rise with the skill, which may reflect the roles drawers are cast in. The NHL's play-by-play feed already records who drew each penalty; putting it on the stat line would cost nothing.

Code and data: github.com/Murray22/extraction-skill

Figure 1 · Out-of-sample validation

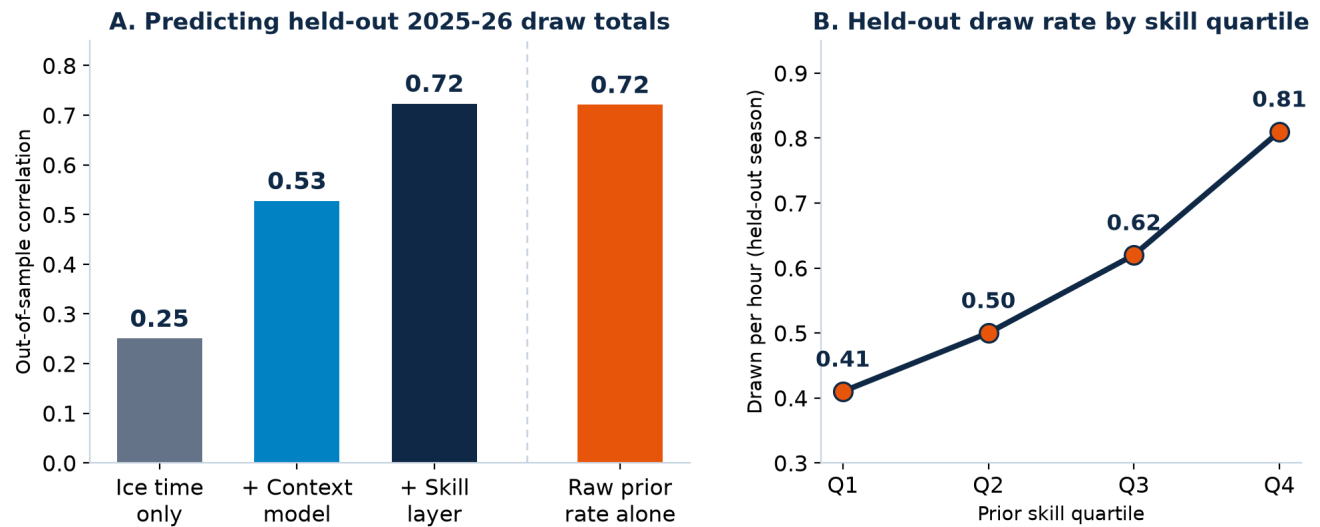


Figure 1. (A) Correlation with held-out 2025–26 player draw totals as model layers are added, beside the player’s raw prior rate on its own. (B) Held-out draw rates by skill quartile, with skill fit on prior seasons only.

Figure 2 · The same method where the answer is already known

LEAGUE	DATA	SPLIT-HALF RELIABILITY	SEASON TO SEASON	TOP OF THE RANKING
NHL	3 seasons · 3,936 games	0.89 (career, 15 seasons)	0.59 mean, 14 pairs	Hathaway 2.45×, Pachal, M. Tkachuk
NBA	5 seasons · 5,830 games	0.94	0.86–0.88, 4 pairs	Antetokounmpo 3.90×, Williamson, Dončić
WNBA	6 seasons · 1,306 games	0.86	0.72–0.82, 5 pairs	Wilson 2.34×, Cambage, Stewart
Soccer, men	4 international tournaments · 230 matches	0.64	–	Hazard 2.98×, Messi, Embolo
Soccer, women	4 international tournaments, 5 club seasons · 949 matches	0.73	–	Milliet 3.89×, Oliviero, Armengol

Figure 2. Drawn fouls are already counted in basketball and soccer, so the method should be stable there and put familiar names on top. **Reliability** compares a player’s rate in two halves of his or her games, scaled to full length (1.0 is perfectly repeatable). **Season to season** is the range of correlations between adjacent seasons; the soccer data are tournaments and single seasons, so it has none. **2.45×** means 2.45 times the rate expected for the position and playing time. NCAA men’s and women’s basketball repeat the result (season to season 0.54 to 0.68) and are in the repository.